

\* Easy

① IP = 10.0.0.5 / 8

class : A

default subnet mask: 255.0.0.0

network bits : 8

host bits : 24

useable hosts per subnet :  $2^n - 2 = 2^{24} - 2$ 

network Address : 10.0.0.0

first useable IP: ~~10.255.255.255~~ 10.0.0.1

Broadcast Address: 10.255.255.255

last useable IP: 10.255.255.254

② IP = 172.16.5.10 / 16

class : B

default subnet mask: 255.255.0.0

network bits : 16

host bits : 16

useable hosts per subnet :  $2^n - 2 = 2^{16} - 2$ 

network Address: 172.16.0.0

first useable IP: 172.16.0.1

Broadcast Address: 172.16.255.255

last useable IP: 172.16.255.254

③ IP = 192.168.10.25 / 24

class : C

default subnet mask: 255.255.255.0

network bits : 24

host bits : 8

useable hosts per subnet:  $2^n - 2 = 2^8$

network Address: 192.168.10.0

first useable IP: 192.168.10.1

broadcast Address: 192.168.10.255

last useable IP: 192.168.10.254

④ IP = 199.1.1.1 / 24

default subnet mask: 255.255.255.0

network bits: 24

host bits: 8

useable hosts per subnet:  $2^8 - 2 = 254$

network Address: 199.1.1.0

first useable IP: 199.1.1.1

broadcast Address: 199.1.1.255

last useable IP: 199.1.1.254

⑤ ① IP = 15.10.5.20

class: A

default subnet mask: 255.0.0.0

② IP = 175.10.40.10

class: B

default subnet mask: 255.255.0.0

③ IP = 210.100.5.1

class: C

default subnet mask: 255.255.255.0

(6)  $IP = 192.168.1.100/24$

network Address:  $192.168.1.0$

first useable IP:  $192.168.1.1$

last useable IP:  $192.168.1.254$

broadcast useable IP:  $192.168.1.255$

(7)  $IP = 172.20.5.0/24$

first useable IP:  $172.20.5.1$

last useable IP:  $172.20.5.254$

(8)  $IP$   $IP$ -class

$5.1.2.3$

A

$130.20.1.1$

B

$192.168.1.1$

C

$230.10.10.1$

D

$250.1.1.1$

E

(9)  $IP: 192.168.50.10/25$

network Address:  $192.168.50.0$

first useable IP:  $192.168.50.1$

broadcast Address:  $192.168.50.127$

last useable IP:  $192.168.50.126$

useable host per subnet:  $2^n - 2$

$$= 2^7 - 2$$

$$= 126$$

⑩ IP: 10.20.30.40 / 16

network Address: 10.20.0.0

broadcast Address: 10.20.255.255

first useable IP: 10.20.0.1

last useable IP: 10.20.255.254

useable Hosts per subnet:  $2^h - 2$   
 $= 2^{16} - 2$

#### \* Medium

① IP = 200.100.50.75 / 26

class = C

default subnet mask: 255.255.255.0

subnet mask: 255.255.255.192 (128)

network bits: 25

host bits: 5

network Address: ~~255.255~~ 200.100.50.0

first useable IP: 200.100.50.1

last useable IP: 200.100.50.126

broadcast Address: 200.100.50.127

② IP: 130.4.102.1 / 22

class: B

default subnet mask: 255.255.0.0

Subnet mask: 255.255.252.0

network bits: 22

host bits: 10

network Address: 130.4.100.0

first useable IP: 130.4.100.1

last useable IP: 130.4.103.254

broadcast addresses: 130.4.103.255

③ IP: 203.1.1.0 / 28

default subnet mask: 255.255.255.0

subnet mask: 255.255.255.240

No. of subnets:  $2^h = 2^{32-28} = 2^4 = 16$

useable host per subnets:  $2^4 - 2 = 14$

first four subnets:

| Subnet | Network    | useable                    | broadcast  |
|--------|------------|----------------------------|------------|
| 1      | 203.1.1.0  | 203.1.1.1 -<br>203.1.1.14  | 203.1.1.15 |
| 2      | 203.1.1.16 | 203.1.1.17 -<br>203.1.1.30 | 203.1.1.31 |
| 3      | 203.1.1.32 | 203.1.1.33 -<br>203.1.1.46 | 203.1.1.47 |
| 4      | 203.1.1.48 | 203.1.1.49 -<br>203.1.1.62 | 203.1.1.63 |

④ IP = 216.21.5.0 / 24

No of hosts per subnet = 30

$$2^h - 2 = 30$$

$$2^h = 32$$

$$h = 5$$

32 - Network bits = Host bits

$$32 - x = 5$$

$$x = 27$$

No of borrows = ~~27~~ 27 - default  
Network bits

No of borrows = 3

new subnet mask = 255.255.255.224

$$\begin{aligned} \text{usable hosts per subnet} &= 2^h - 2 \\ &= 2^5 - 2 \\ &= 30 \end{aligned}$$

$$\text{no of subnets} = 2^{\text{borrowed bits}} = 2^3 = 8$$

first two Subnet blocks:

$$\text{block size} = 256 - 224 = 32$$

$$\begin{array}{r} 1110 \ 0000 \\ 128 + 64 + 32 = 224 \end{array}$$

| Subnet | Network     | usable Range                 | broadcast                             |
|--------|-------------|------------------------------|---------------------------------------|
| 1      | 216.21.5.0  | 216.21.5.1 -<br>216.21.5.30  | <del>216.21.5.31</del><br>216.21.5.31 |
| 2      | 216.21.5.32 | 216.21.5.33 -<br>216.21.5.62 | 216.21.5.63                           |

⑤ IP = 192.168.100.0 / 26

No of subnets =  $2^{\text{borrowed bits}} = 2^2 = 4$

useable hosts per subnet =  $2^h - 2$   
 $= 2^6 - 2$   
 $= 62$

block size =  $256 - 192$  (1100 0000)  
 $= 64$  128 + 64

| Subnet | Network         | usable range                         | broadcast       |
|--------|-----------------|--------------------------------------|-----------------|
| 1      | 192.168.100.0   | 192.168.100.1 -<br>192.168.100.62    | 192.168.100.63  |
| 2      | 192.168.100.64  | 192.168.100.65 -<br>192.168.100.126  | 192.168.100.127 |
| 3      | 192.168.100.128 | 192.168.100.129 -<br>192.168.100.190 | 192.168.100.191 |

⑥ IP1 = 172.16.17.30 / 20  
 IP2 = 172.16.28.15 / 20

block size =  $256 - 240$  (1111 0000)  
 $= 16$  240

0-15, 16-31, 32-47  
 ↓

both will be in one subnet

⑦  $182.44.82.16/26$

Network Address:  $182.44.82.0$

broadcast Address:  $182.44.82.63$

Useable range:  $182.44.82.1 - 182.44.82.62$

⑧  $IP1 = 192.168.10.25/23$

$IP2 = 192.168.11.200/23$

block size =  $256 - 254 = 2$

0-1, 2-3, 4-5

10 & 11

Both will be in  
Same subnet.

Network Address =  $192.168.10.0$

⑨  $IP = 172.16.10.100/20$

subnet mask =  $255.255.240.0$

useable host per subnet =  $2^{\text{borrow}} - 2$   
 $= 2^{12} - 2 = 4096 - 2 = 4094$

network Address =  $172.16.0.0$

broadcast Address =  $172.16.15.255$

first useable Address =  $172.16.0.1$

last useable Address =  $172.16.15.254$

(10) ~~Attempt~~ IP = 192.168.10.0/24  
 No of subnets = 14

$$2^{\text{host}} \geq 14$$

$$2^4 \geq 14$$

$$\text{host} = 4 \rightarrow 2^4 \rightarrow 16 \text{ subnets}$$

↳ 2 borrowed bits

borrow bits : 4

new subnet mask : 255.255.255.240  
 (128)

no of subnets : 16

$$\begin{aligned} \text{usable host per subnet} &= 2^h - 2 \\ &= 2^4 - 2 \\ &= 14 \end{aligned}$$

\* Hard

(1) IP = 10.10.0.0/16

$$\begin{aligned} \text{No of users per subnet} &\geq 500 \\ 2^h &\geq 502 \end{aligned}$$

$$\text{host bit} = 9 \rightarrow \text{since 9 bits are required so minimum prefix length}$$

$$\text{Network bit} = 32 - 9 = 23 = 23$$

$$\text{Subnet mask} = 255.255.254.0 \quad \{32-9\}$$

$$\text{borrowed bit} = 23 - 16 = 7$$

$$\text{useable host per subnet} = 2^{\text{borrowed}} - 2$$

$$= 2^7 - 2 = 126$$

$$\text{block size} = 256 - 254 = 2$$

$$\text{no of equal subnet} = 2^{\text{borrowed}} = 2^7 = 128$$

| Subnet | Network Address | useable IP Range        | Broadcast   |
|--------|-----------------|-------------------------|-------------|
| 1      | 10.10.0.0/23    | 10.10.0.1 - 10.10.1.254 | 10.10.1.255 |
| 2      | 10.10.2.0/23    | 10.10.2.1 - 10.10.3.254 | 10.10.3.255 |
| 3      | 10.10.4.0       | 10.10.4.1 - 10.10.5.254 | 10.10.5.255 |
| 4      | 10.10.6.0       | 10.10.6.1 - 10.10.7.254 | 10.10.7.255 |

2 IP = 172.20.0.0/16

$$2^h - 2 \geq 1000$$

$$2^h \geq 1002$$

$$h \geq 10 \rightarrow \text{Host bit} \geq 10$$

$$\text{minimum prefix length} = \text{network bit}$$

$$= 32 - 10 = 22$$

$$\text{Subnet mask} = 255.255.252.0$$

$$\text{borrowed bits} = 6$$

$$\text{No of subnets} = 2^{\text{borrowed}} = 2^6 = 64$$

first four subnets:

$$\text{block size} = 256 - 252 = 4$$

↳ 3rd octet increased by 4

$$1\text{st: } 172.20.0.0 / 22$$

$$2\text{nd: } 172.20.4.0 / 22$$

$$3\text{rd: } 172.20.8.0 / 22$$

$$4\text{th: } 172.20.12.0 / 22$$

③  $\text{IP} = 150.10.0.0 / 16$

$$2^h - 2 \geq 250$$

$$2^h \geq 252$$

$$h \geq 8$$

$$\text{minimum subnet prefix} = 32 - 8 = /24$$

$$\text{Minimum subnet size} = 2^{\text{host}} = 2^8 = 256$$

$$\text{Subnet mask} = 255.255.255.0$$

$$\text{No of available subnets} = 2^8 = 256$$

used subnets = Available - used.

$$= 256 - 100 \text{ (dept)}$$

$$= 156$$

useable hosts per subnet =  $2^8 - 2$

$$= 254$$

first four Subnet Address:

1.  $150.10.0.0 / 24$

2.  $150.10.1.0 / 24$

3.  $150.10.2.0 / 24$

4.  $150.10.3.0 / 24$

IP =  $172.31.100.0 / 21$

broadcast Address =  $172.31.103.255$

~~Subnet~~  $172.31.96.0$

Network Address =  $172.31.96.0$

useable range =  $172.31.96.1 - 172.31.103.254$

useable hosts per subnet = ~~32~~  $2^8 - 2$

No of  $/21$  subnets from original class B  $\rightarrow /16$

= 2 borrowed =  $2^5 = 32$

5)  $IP = 11.0.0.0 / 8$

$$2^h - 2 \geq 2046$$

$$2^h \geq 2048$$

$$h \geq 11$$

$$\begin{aligned} \text{minimum prefix length} &= \text{Network bits} \\ &= 32 - 11 \\ &= /21 \end{aligned}$$

$$\text{Subnet mask} = 255.255.248.0$$

$$\text{Borrowed bits} = 13$$

$$\text{No of subnets} = 2^{13}$$

$$\begin{aligned} \text{first four subnet Address} \\ &= 256 = 248 = 8 \end{aligned}$$

$$1. \quad 11.0.0.0 / 21$$

$$2. \quad 11.0.8.0 / 21$$

$$3. \quad 11.0.16.0 / 21$$

$$4. \quad 11.0.24.0 / 21$$

6)  $IP = 100.10.0.0 / 8$

$$2^h - 2 \geq 1200$$

$$2^h \geq 1202$$

$$h \geq 11 \rightarrow \text{minimum prefix} = 32 - 11 = /21$$

net mask = 255.255.248.0

borrowed bits = 5 13

useable hosts per subnet =  $2^h - 2$   
 $= 2^{11} - 2$   
 $= 2046$

No of equal subnets =  $2^{13}$

first five subnet addresses :  
 $256 - 248 = 8$

1) 100.0.0.0 /21

2) 100.0.8.0 /21

3) 100.0.16.0 /21

4) 100.0.24.0 /21

5) 100.0.32.0 /21

IP = 192.168.40.0 /24

TV = 100 + 50 + 25 = 175

Dept: A  $2^h - 2 \geq 100$

$2^h \geq 102$

$h \geq 7 \rightarrow /25$

Dept: B  $2^h - 2 \geq 50$

$2^h \geq 52$

$h \geq 6 \rightarrow /26$

Dept: C  $2^h - 2 \geq 25$

$2^h \geq 27$

$h \geq 5 \rightarrow /27$

Subnet mask A  $\rightarrow 255.255.255.128$   
 B  $\rightarrow 255.255.255.192$   
 C  $\rightarrow 255.255.255.224$

Network Address A  $\rightarrow 192.168.40.0$   
 B  $\rightarrow 192.168.40.128$   
 C  $\rightarrow 192.168.40.192$

Useable range A  $\rightarrow 192.168.40.1 - 192.168.40.127$   
 B  $\rightarrow 192.168.40.129 - 192.168.40.190$   
 C  $\rightarrow 192.168.40.193 - 192.168.40.223$

broadcast  $\rightarrow$  A  $\rightarrow 192.168.40.127$   
 B  $\rightarrow 192.168.40.191$   
 C  $\rightarrow 192.168.40.223$

IP<sub>1</sub> = 172.16.70.130 / 18

IP<sub>2</sub> = 172.16.130.200 / 18

subnet mask = 255.255.192.0

block size = 256 - 192 = 64

3rd

0 - 63

64 - 127  $\rightarrow$  IP<sub>1</sub>

128 - 191  $\rightarrow$  IP<sub>2</sub>

IP<sub>1</sub> Network address = 172.16.<sup>64</sup>~~128~~.0

broadcast Add = 172.16.127.255

range = 172.16.64.1 - 172.16.127.254

IP<sub>2</sub> Network Address = 172.16.128.0

broadcast Add = 172.16.191.255

range = 172.16.128.1 - 172.16.191.254

$$IP = 192.168.1.0 / 24$$

(10)

$$2^h - 2 \geq 30$$

$$2^h \geq 32$$

$$\underline{h \geq 5} \rightarrow \text{Network bits} = 32 - 5 = \underline{27}$$

$$\text{subnet mask} = 255.255.255.224.$$

$$\text{no of subnets} = 2^{\text{borrow}} = 2^3 = 8$$

$$\begin{aligned} \text{usable host per subnets} &= 2^h - 2 \\ &= 2^5 - 2 \\ &= \underline{30} \end{aligned}$$

$$\text{All subnet blocks} = 256 - 224 = 32$$

$$192.168.1.0 / 27$$

$$192.168.1.32 / 27$$

$$192.168.1.64 / 27$$

$$192.168.1.96 / 27$$

$$192.168.1.128 / 27$$

$$192.168.1.160 / 27$$

$$192.168.1.192 / 27$$

$$192.168.1.224 / 27$$

8 equal size subnets

FLSM  $\rightarrow$  fixed length subnet marking

In Here we have 8 subnets

we need only 6.

In yes, possible.